Simple Linear Regression

Lecture 14

Topics in this lecture

- Correlation
- Simple linear regression

Correlation

- Frequently, we would like to describe a relationship between 2 variables in a data set
- Examples
  - What's the relation between ages of Facebook users and the number of friends they have?
  - What's the relation between height and arm length of a person? Does a taller person tend to have a longer arm length?

Correlation Coefficient

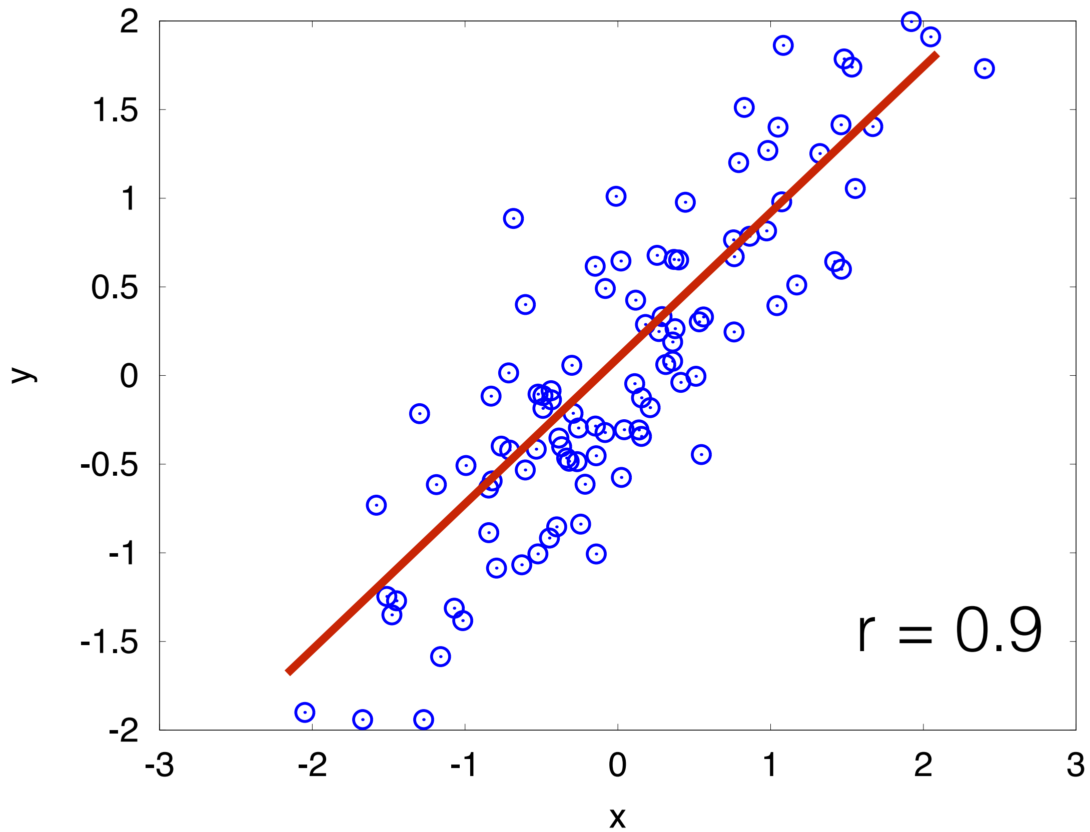
- Correlation coefficient, denoted by ***r***, is a quantity describing how closely related two variables are.

- The value of r indicates the strength of the linear relationship between two variables

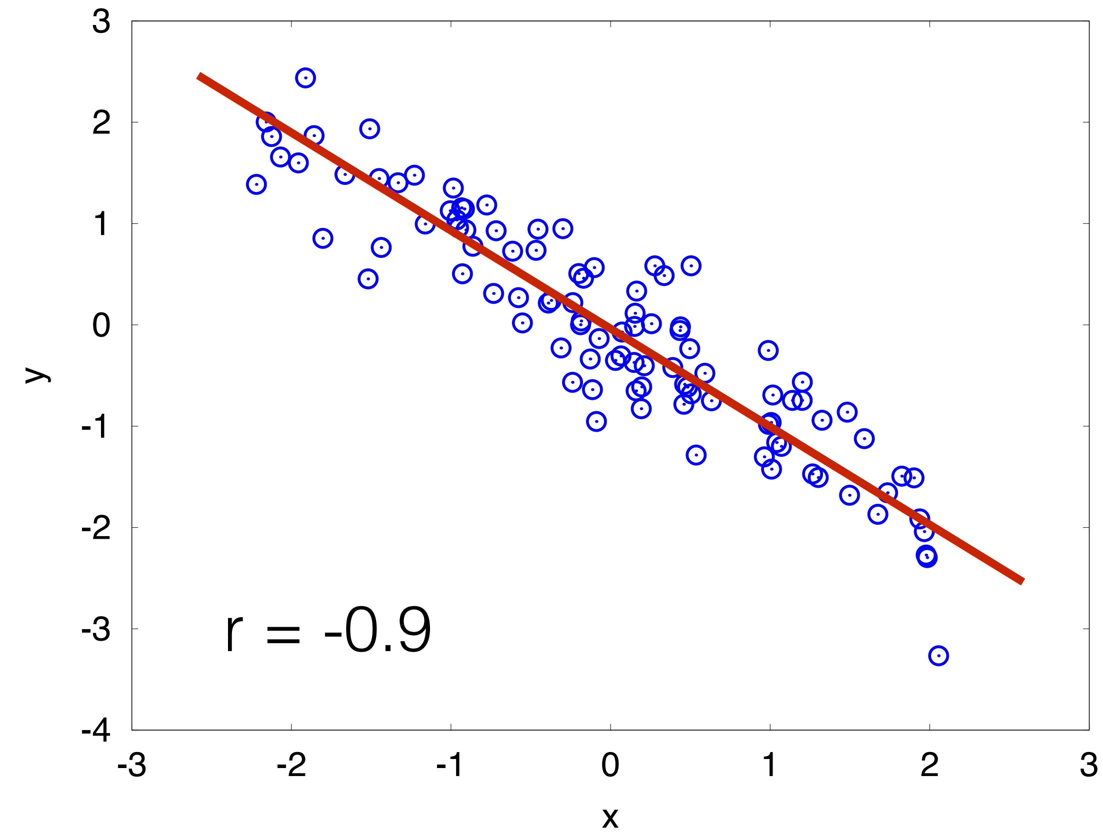
- The range of r is from -1 to 1

- If $r = 0$, we say that the two variables are uncorrelated

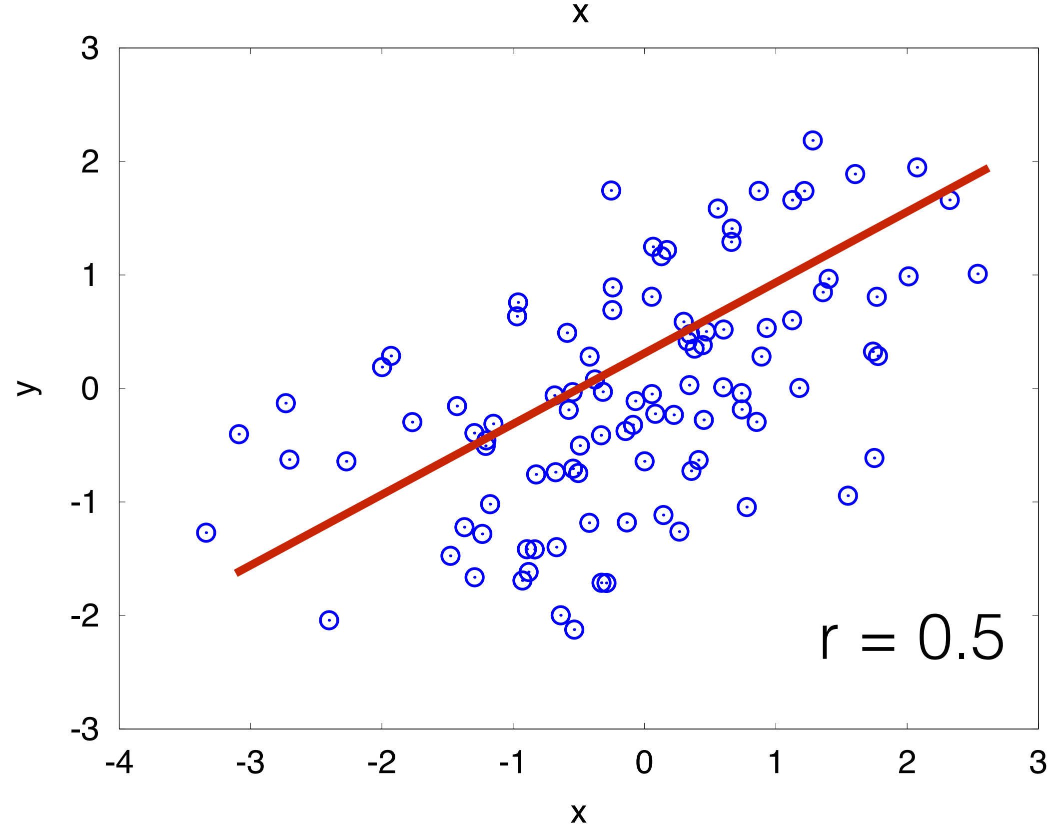
Values of r



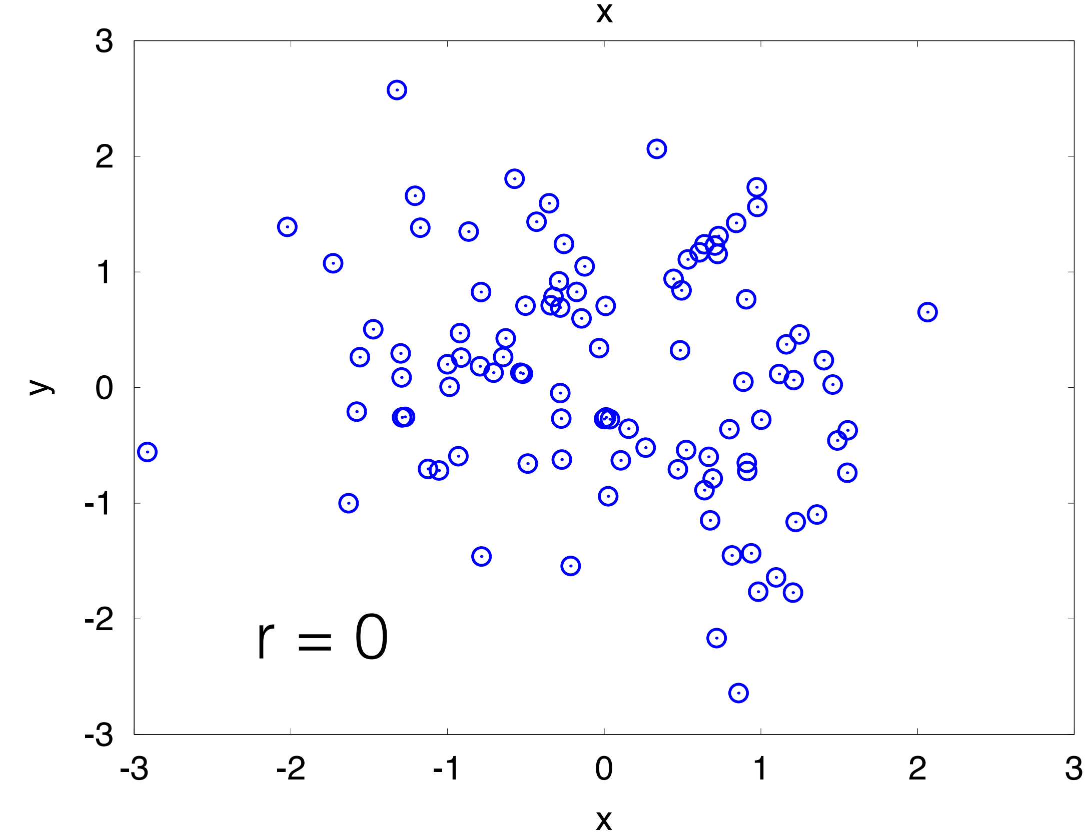
r = 0.9
y
x
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-3
-2
-1
0
1
2
3



r = -0.9
y
x
3
2
1
0
-1
-2
-3
-4
-3
-2
-1
0
1
2
3



r = 0.5
y
x
3
2
1
0
-1
-2
-3
-4
-3
-2
-1
0
1
2
3



r = 0
y
x
3
2
1
0
-1
-2
-3
-3
-2
-1
0
1
2
3

Calculating *r*

- Given two variables x,y in the data set, the correlation coefficient can be computed as

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i - \overline{x}}{s_x}\right)\left(\frac{y_i - \overline{y}}{s_y}\right)$$

- Alternatively

$$r = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \overline{y})^2}}$$

$$r = \frac{\sum_{i=1}^{n} x_i y_i - n\overline{x}\,\overline{y}}{\sqrt{\sum_{i=1}^{n} x_i^2 - n\overline{x}^2}\sqrt{\sum_{i=1}^{n} y_i^2 - n\overline{y}^2}}$$

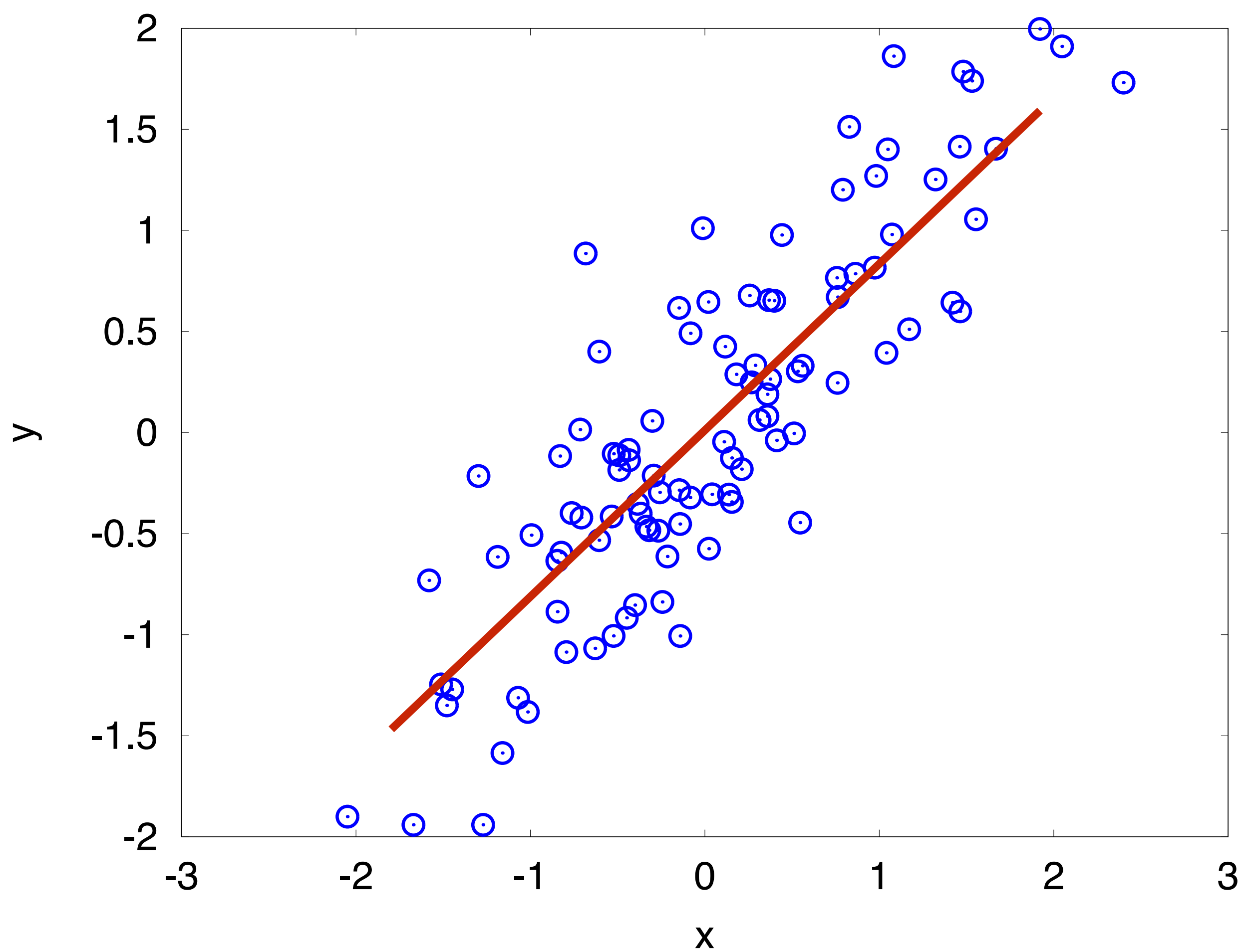
Example 1

- Given two variables x,y in the table. Find the correlation coefficient

x	y
2	4
3	8
7	12
4	9
9	15

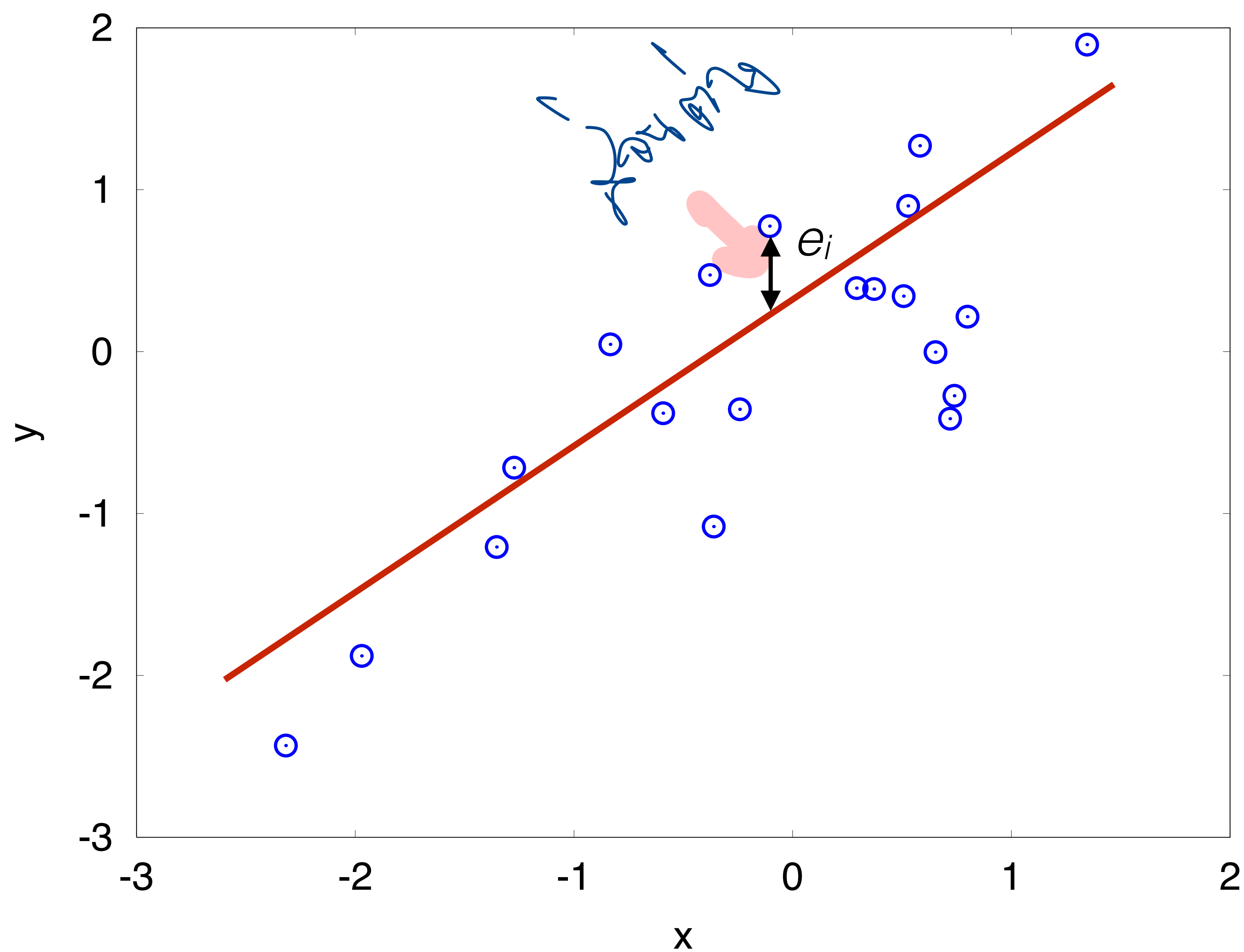
Linear Regression

- Main idea
  - We have 2 variables (x,y)
  - We want to find a line that best fit these data



Least Square Line

- A line that best fit the data $\longrightarrow$ minimize $\sum_{i=1}^{n} e_i^2$



Regression Line

- Actual equation describing a relation between y_i and x_i

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

- Estimated equation

$$y = \widehat{\beta_0} + \widehat{\beta_1} x$$

Best Line

- Residual term

$$e_i = y_i - \widehat{y}_i \quad = \quad y_i - \widehat{\beta}_0 - \widehat{\beta_1} x_i$$

- Want to minimize

$$S = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \widehat{\beta}_0 - \widehat{\beta_1} x_i)^2$$

- Solve the following system of equations

$$\frac{\partial S}{\partial \widehat{\beta}_0} = -\sum_{i=1}^{n} 2(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i) \quad = \quad 0$$

$$\frac{\partial S}{\partial \widehat{\beta}_1} = -\sum_{i=1}^{n} 2x_i(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i) \quad = \quad 0$$

Best Line (2)

- Least square line

$$y = \widehat{\beta_0} + \widehat{\beta_1} x$$

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

$$\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$$

Example 2

- Given two variables x,y in the table. Find the regression line

x	y
2	4
3	8
7	12
4	9
9	15